

Diagnosing Conformal Prediction Failures Under Distribution Shift

A COVID-19 Case Study

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KEY FINDING: SHAP Concentration $(\varphi_1/\sum|\varphi_j|)$ predicts CP failure with $\rho = 0.853$ | 40% threshold for vulnerability

Problem & Motivation

Conformal Prediction (CP) provides distribution-free uncertainty with **guaranteed coverage**.

The Challenge:

- Real-world data has **distribution shift**
- COVID-19: **extreme temporal shifts**
- CP guarantees **break down**

Research Question:

Can we **predict** when CP fails **before** deployment?

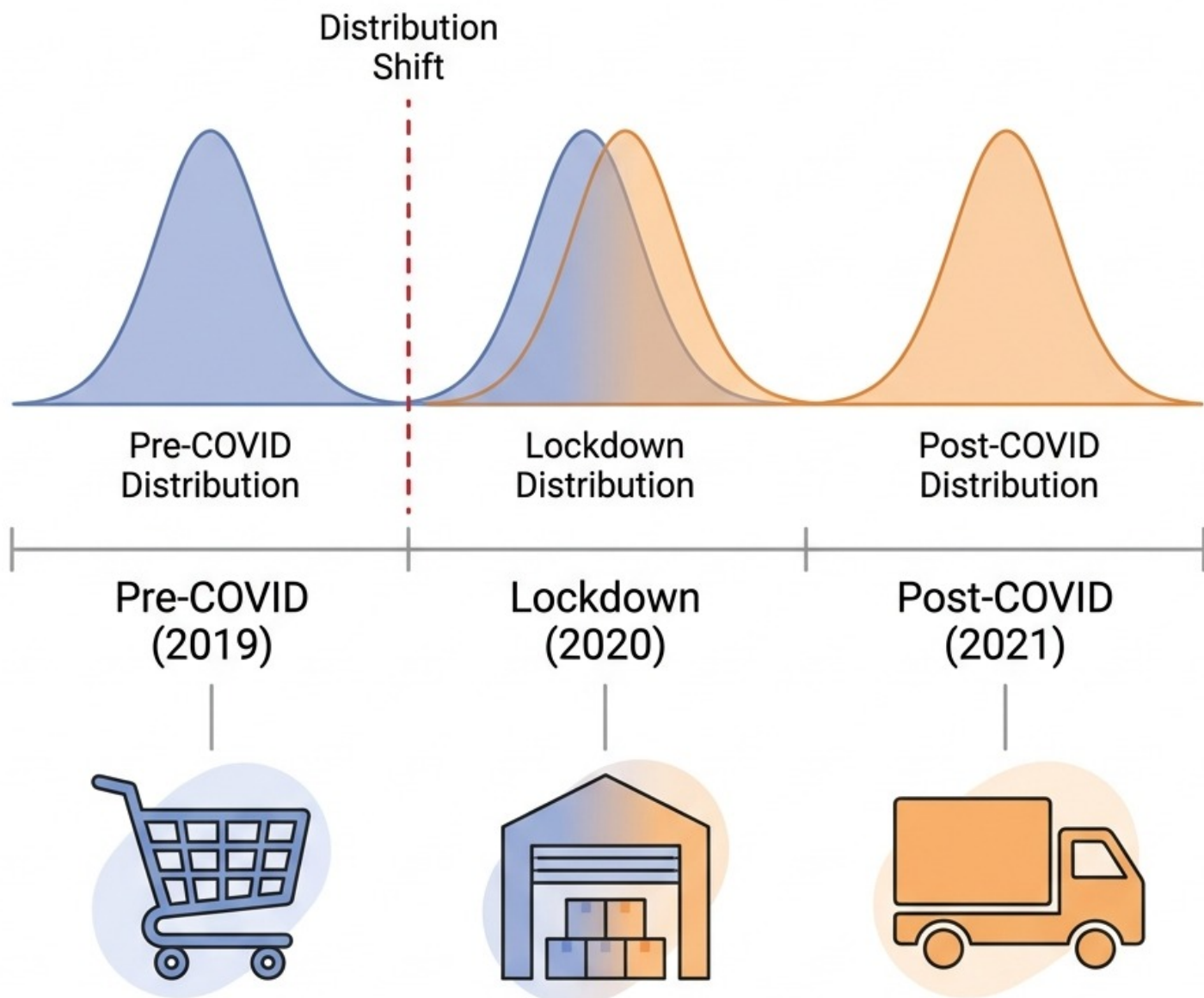


Fig 1: COVID-19 distribution shifts

SHAP Concentration Method

Definition:

$$\text{SHAP Conc.} = \frac{\varphi_1}{\sum_{j=1}^p |\varphi_j|}$$

φ_1 = Top feature SHAP; $\sum |\varphi_j|$ = Total SHAP

Decision Threshold (40%):

< 40%
Robust

≥ 40%
Vulnerable

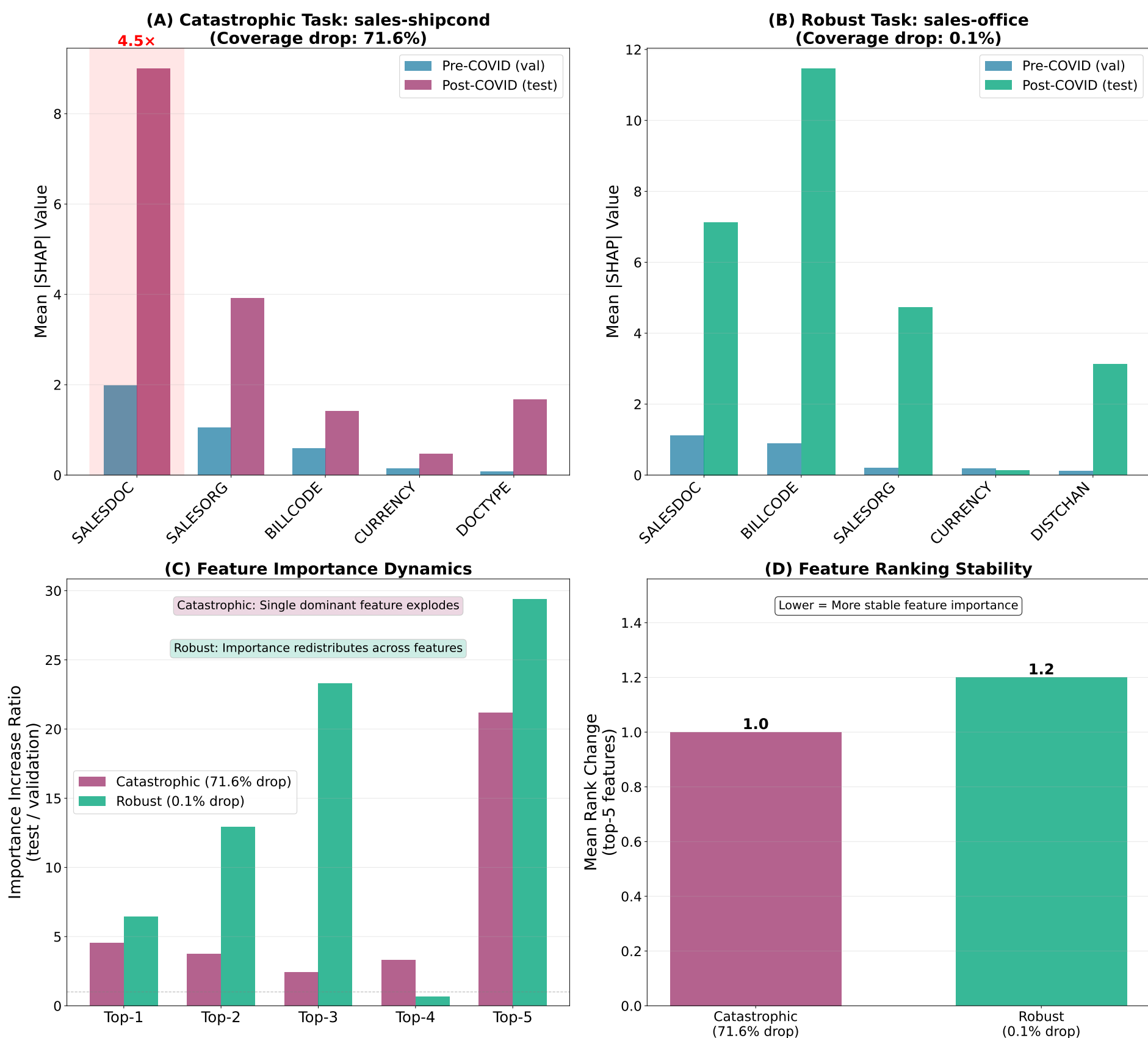


Fig 2: Feature importance concentration

External Validation: Covertypes

UCI Forest Covertypes Dataset

- 581K samples, 54 features
- Geographic distribution shift

81.8pp
Coverage Drop
90% → 49.8%

Validates 40% threshold across domains

Key Insight

Models with **concentrated feature reliance** are vulnerable:

- Single feature drift ⇒ large impact
- No redundant features for backup
- Calibration misalignment

High Concentration

↓
High Vulnerability

SALT COVID Dataset

Clinical Trials Database

- 9 clinical domains
- 16 prediction tasks
- Pre → pandemic shift

Domain	Tasks
Cardiovascular	3
Respiratory	2
Emergency/ICU	4
Lab/Mortality	4
Other	3

MAIN RESULTS

Strong Correlation: SHAP Concentration vs Coverage Drop

$$\rho = 0.853$$

Spearman Correlation

$p < 0.001$ | $n = 16$ tasks | 9 domains

SALT COVID (n=8): $\rho = 0.833$ ($p = 0.010$)

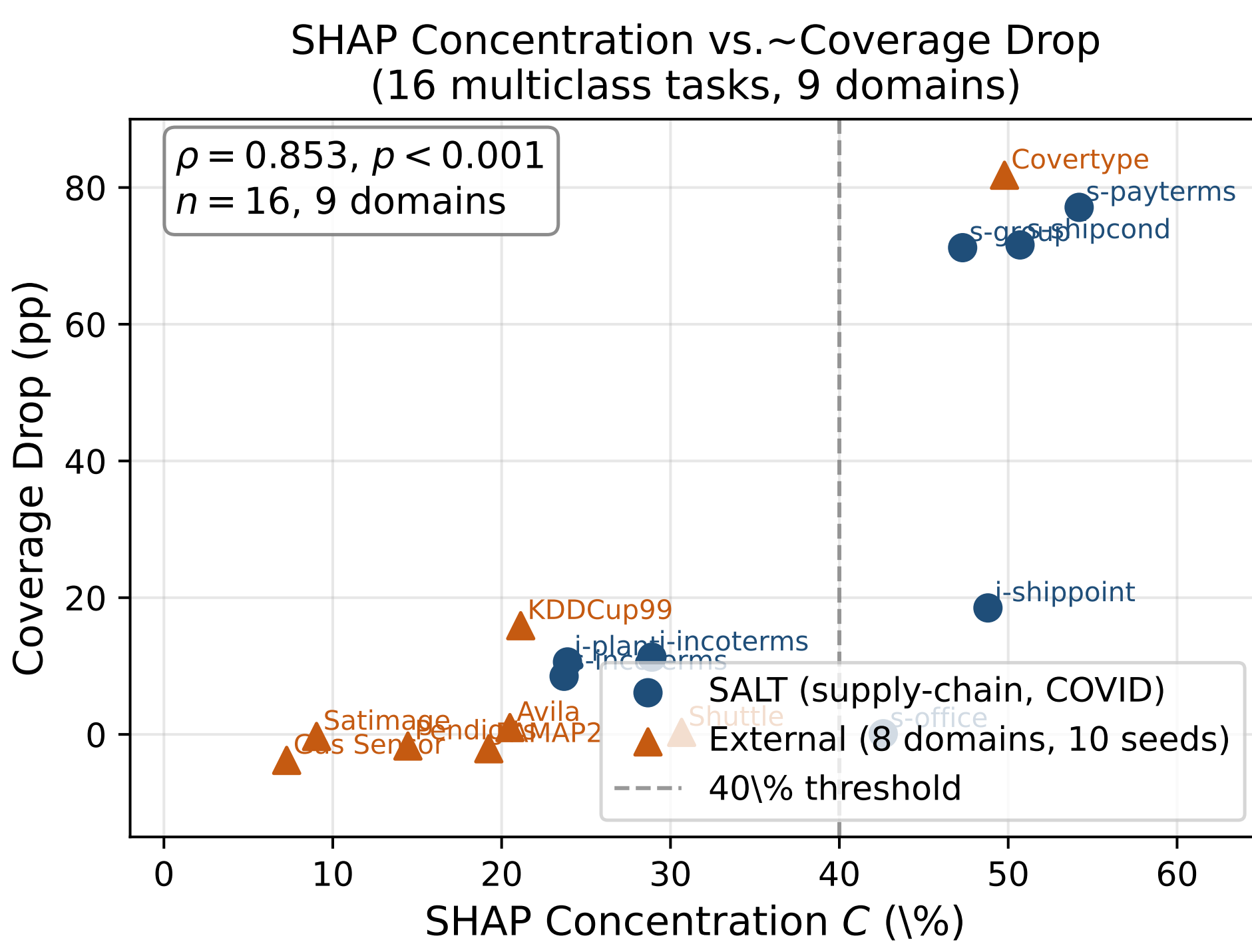
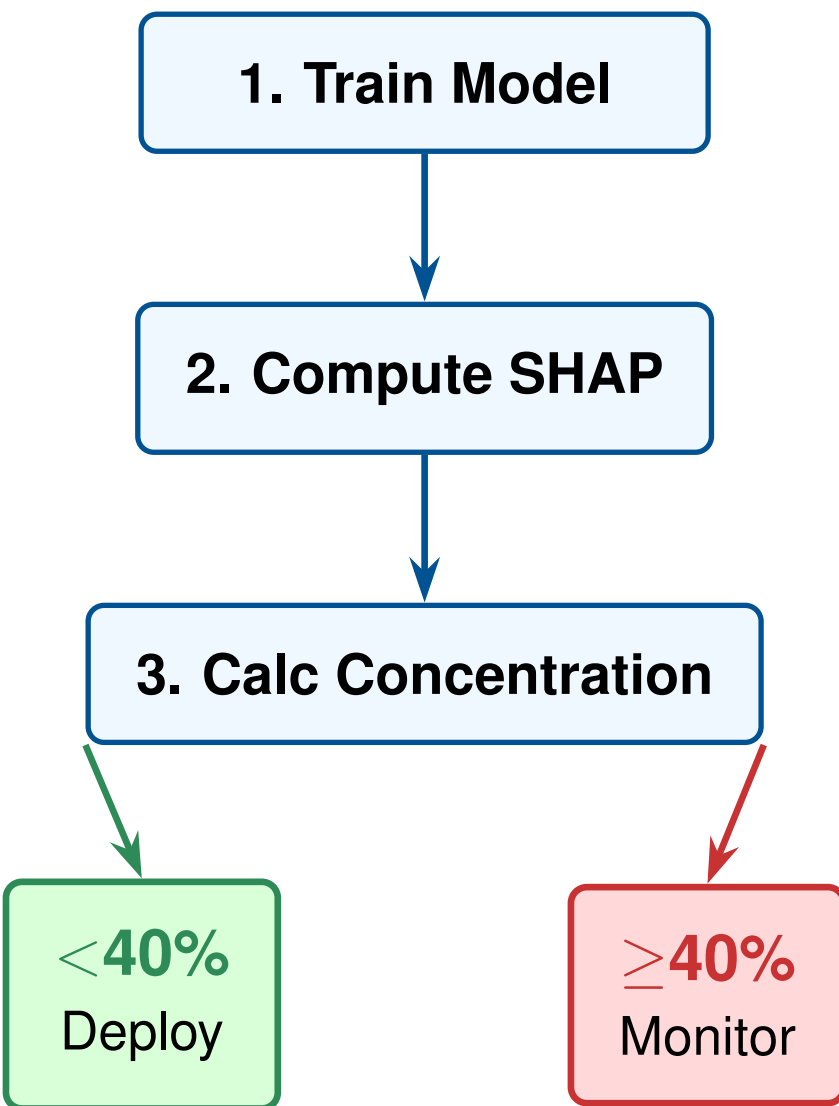


Fig 3: SHAP Concentration vs Coverage Drop (n=16)

Practical Framework

Pre-Deployment Assessment



Conclusions

Contributions:

- SHAP Concentration predicts CP failures
- Strong correlation: $\rho = 0.853$ ($p < 0.001$)
- Practical 40% threshold
- Validated on Covertypes dataset

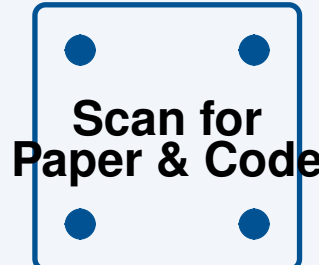
Future: Adaptive CP, real-time monitoring, DL extension

Paper & Code:

github.com/chorok-lee/conformal-covid

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Model Comparison

Model	ρ	Rank
LightGBM	0.833	1
CatBoost	0.786	2
XGBoost	0.762	3
Random Forest	0.714	4